

Discovery Piscine - AI/ML

Module 3 - Genetic Algorithms

Summary: You will learn to use genetic algorithms to solve optimization problems by simulating natural evolution.

Version: 1.01

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Chapter I

General instructions

Unless explicitly specified, the following rules will apply to this project.

- This subject is the one and only trustable source. Don't trust any rumor.
- Your assignments will be evaluated by your peers.
- You have a question? Ask your left neighbor. Otherwise, try your luck with your right neighbor.
- You must read the examples thoroughly. They can reveal requirements that are not obvious in the assignment's description.
- Good luck and good work on this project!

Chapter II

Exercise 0: Genetic Algorithms

	Exercise : 00
Exercise 0: Genetic Algorithms	
Turn-in directory : $ex00/$	
Files to turn in : `genetic_algorithms.py`	
Forbidden functions : `None`	

In this project, you will create a tool using Genetic Algorithms to determine the optimal frequency settings for antennas placed around Forty2.

Forty2 is a historic fort on the west coast of France, where students and alumni can spend time during summer. With dormitories and a cluster, it's the perfect place to relax and code, nestled in a beautiful pine forest.

Unfortunately, the 4G/5G cellular connection is suboptimal (don't worry, the WiFi works fine!). Depending on your location within the fort, cellular reception can be poor. Your job is to optimize it!
This involves creating a program to optimize antenna frequencies using a genetic algorithm. Users will adjust frequencies within a specific range to achieve the best balance and overall connection across all areas. Five key locations in the fort will be the focus of this exercise: the dorms, the kitchen, the showers, the cluster and the entrance.



The genetic algorithm will begin with multiple random frequency settings (known as the initial population), evaluate them using a fitness function, select the best-performing ones, and apply crossover and mutations to generate new settings (a new generation of the population). It will then display the best settings for each generation. After several generations, your algorithm will identify the best possible combination.

For this project, follow these steps:

- `Generate Initial Population`:
  - Generate 1000 individuals, each with 4 random frequency values between 45.0 and 50.0. These individuals represent potential combinations of antenna frequency settings.
  - Example: For each of the 1000 individuals, generate four random frequency values within the specified range.

- `Evaluate Each Individual`:
  - Pass each individual through the provided binary to calculate 5 reliability indices (ranging approximately between 0 and 5), corresponding to the 5 locations: dorms, kitchen, showers, cluster and entrance.
  - Example: Each individual will produce different indices for the locations, reflecting the quality of the antennas mix.

```
./cellular 46.26 49.89 45.12 47.47
Cellular quality
 dorms    : 0.035
 kitchen  : 2.185
 showers  : 1.166
 cluster  : 4.028
 entrance: 0.290
```

- `Selection`:
  - Select the frequency settings from the current population to retain for the next generation, based on their fitness scores.
  - Example: Identify individuals with high performance across multiple locations or strong scores in specific areas.

- `Crossover and Mutation`:
  - Implement crossover and mutation functions to generate new individuals for the next generation of frequency settings.
  - Example: Combine high-performing individuals to see if intermediate values yield better results. Replace poor performers with new random values or eliminate them entirely.

- `Iterate`:
  - Repeat the process for multiple generations while maintaining a stable population size.
  - Example: Iterate 100, 1000, or 10000 times, ensuring the population doesn't drop below a threshold or grow excessively. Add random individuals if the population shrinks too much, or trim it if it exceeds a set size.

- `Display Best Settings`:
  - Display the best frequency settings from each generation to observe overall optimization progress.
  - After completing all iterations, display the best settings found, which should be among the final generations.
  - Example: Show the best individuals and their corresponding reliability indices to visualize the optimization process.

`objective`:

- Create a tool using a genetic algorithm to optimize antenna frequency settings.
- Allow users to adjust parameters such as the number of iteration, population size, mutation strategy, and more.
- Observe how the genetic algorithm refines frequency settings based on these adjustments.

Chapter III

Submission and peer-evaluation

- Create an `ai-ml_discovery` folder at the root of your home, and move around in it.

- Create a new module folder with this format:

```
module[number_of_the_module]
example: module3 (for module 3)
```

  and navigate to it.

- From now on, all exercises should be in the correct folder rendering. Exercise 00 in the `ex00` folder, Exercise 01 in the `ex01` folder, etc.
  for example:

```
[module3/ex00]
[module3/ex01]
etc...
```



Please note, during your defense anything that is not present in the folder for the day will not be checked.